DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-D13: The Dairy farm Case [20 marks]

Briefing: You are the data detective hired by the manager of a dairy farm. The file holds 14 monthly records from the monsoon year, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
rainfall	Rainfall (mm)	crop_damage	Crop damage (%)
grass_height	Grass height (cm)	tractor_hours	Tractor hours
milk_yield	Milk yield (litres per cow)	fuel_cost	Fuel cost (thousand rupees)
fert_dev	Fertiliser minus recommended dose (kg per ha)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(132578)
n <- 14
rainfall <- round(rnorm(n, 124.6, 39.3), 1)
grass_height <- round(36.3 + 8.8 * (0.94 * (rainfall - 124.6) / 39.3 + 0.35 * rnorm(n)), 1)
milk_yield <- round(14.5 + 2.9 * (-0.89 * (rainfall - 124.6) / 39.3 + 0.46 * rnorm(n)), 1)
fert_dev <- round(seq(-32.8, 32.8, length.out = n)[order(runif(n))], 1)
crop_damage <- round(6.2 + 21.4 * (fert_dev / 32.8)^2 + rnorm(n, 0, 1.1), 1)
tractor_hours <- round(rnorm(n, 93, 19.7), 0)
fuel_cost <- round(46.7 + 9.8 * (0.07 * (tractor_hours - 93) / 19.7 + 1.00 * rnorm(n)), 1)
tractor_hours[5] <- 158; fuel_cost[5] <- 77.2   # row 5 was typed in by a trainee
case <- data.frame(rainfall, grass_height, milk_yield, fert_dev, crop_damage, tractor_hours, fuel_cost)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot crop_damage against fert_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot fuel_cost against tractor_hours. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. grass_height and milk_yield are strongly linked. Report r(grass_height, milk_yield), then r(rainfall, grass_height) and r(rainfall, milk_yield). Compute the partial correlation of grass_height and milk_yield controlling for rainfall: correlate resid(lm(grass_height ~ rainfall)) with resid(lm(milk_yield ~ rainfall)). Write a two-sentence verdict: does grass_height cause milk_yield? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

Question 1: Calculate the Correlation Matrix (4 Marks)

Aim

To calculate the correlation matrix for seven dairy farm variables using R and identify the number of observations and distinct pairs of variables.

Theory

A correlation matrix shows the relationship between variables. The correlation coefficient ranges from -1 to +1. A positive value indicates that two variables tend to increase together, while a negative value indicates that one variable tends to decrease when the other increases.

Calculation

Number of observations:

$$n=14$$

Number of variables:

$$ p=7 $$

Number of distinct pairs:

$$ \frac{p(p-1)}{2} =\frac{7(7-1)}{2} =\frac{42}{2}=21 $$

Therefore, there are 14 observations, 7 variables, and 21 distinct pairs of variables.

R Program

```
# Calculate correlation matrix
cor_matrix <- cor(case)

# Display correlation matrix rounded to 2 decimals
round(cor_matrix, 2)

# Count observations and variables
n <- nrow(case)
p <- ncol(case)

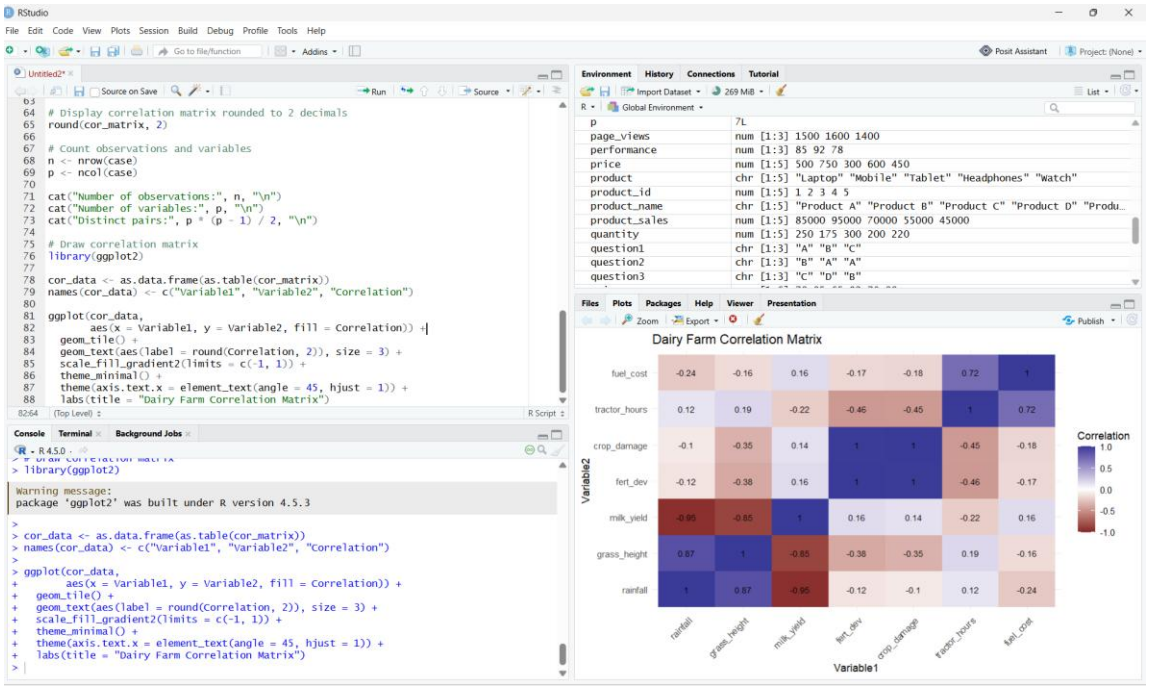
cat("Number of observations:", n, "\n")
cat("Number of variables:", p, "\n")
cat("Distinct pairs:", p * (p - 1) / 2, "\n")

# Draw correlation matrix
library(ggplot2)

cor_data <- as.data.frame(as.table(cor_matrix))
names(cor_data) <- c("Variable1", "Variable2", "Correlation")

ggplot(cor_data,
     aes(x = Variable1, y = Variable2, fill = Correlation)) +
  geom_tile() +
```

```
geom_text(aes(label = round(Correlation, 2)), size = 3) +
scale_fill_gradient2(limits = c(-1, 1)) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
labs(title = "Dairy Farm Correlation Matrix")
```



Result

The correlation matrix contains 7 rows and 7 columns. The dataset has 14 observations and 21 distinct pairs of variables. The correlogram visually represents the positive and negative relationships between the variables.

Question 2: Identify the Strongest Correlation Clues (3 Marks)

Aim

To identify the strongest positive and strongest negative correlations among the dairy farm variables.

Theory

Correlation measures the strength and direction of a linear relationship.

- Weak correlation: absolute value below 0.30.
- Moderate correlation: absolute value from 0.30 to 0.70.
- Strong correlation: absolute value above 0.70.

R Program

```
# Calculate the correlation matrix
cor_matrix <- cor(case)

# Remove diagonal values
diag(cor_matrix) <- NA

# Find strongest positive correlation
max_position <- which(
  cor_matrix == max(cor_matrix, na.rm = TRUE),
  arr.ind = TRUE
)[1, ]

# Find strongest negative correlation
min_position <- which(
  cor_matrix == min(cor_matrix, na.rm = TRUE),
  arr.ind = TRUE
)[1, ]

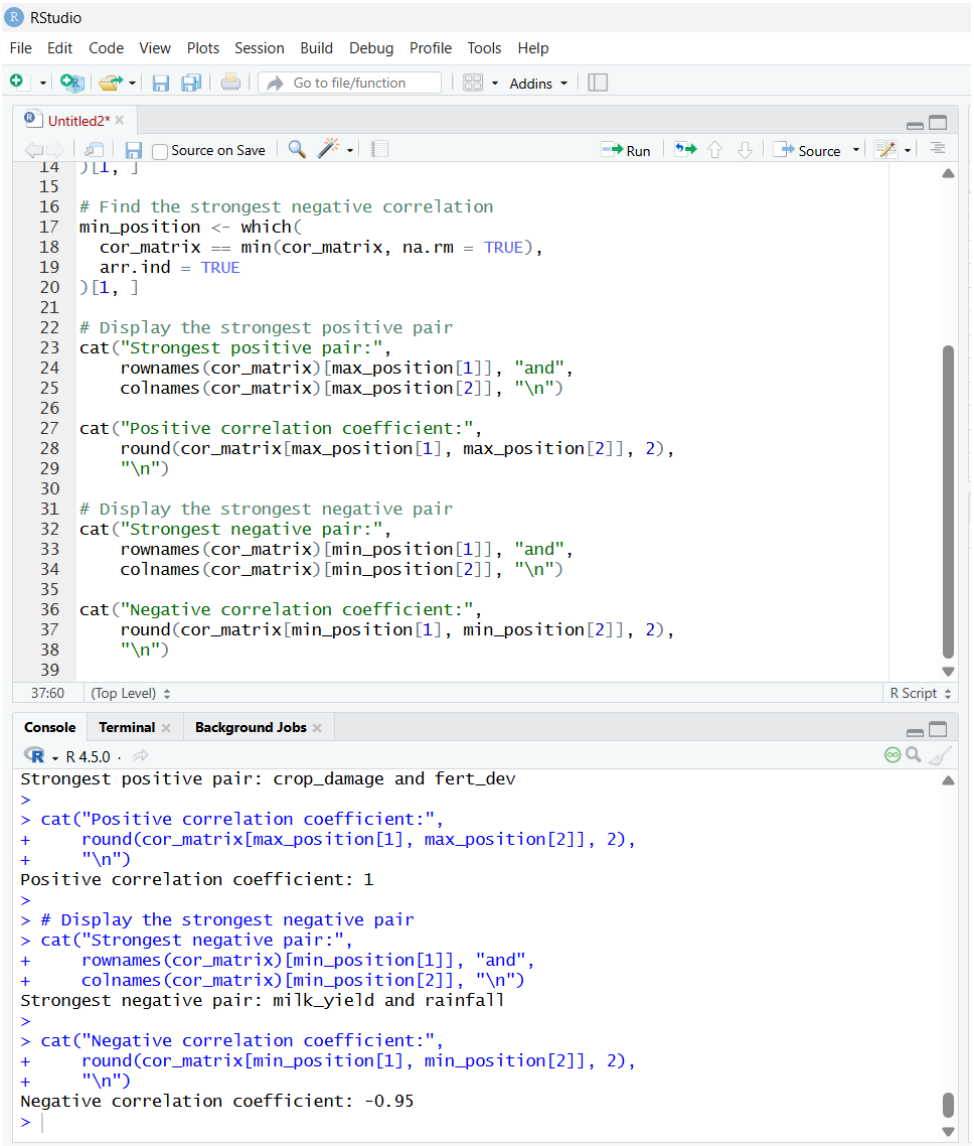
cat("Strongest positive pair:",
    rownames(cor_matrix)[max_position[1]], "and",
    colnames(cor_matrix)[max_position[2]], "\n")

cat("Positive correlation:",
    round(cor_matrix[max_position[1], max_position[2]], 2), "\n")

cat("Strongest negative pair:",
    rownames(cor_matrix)[min_position[1]], "and",
```

```
  colnames(cor_matrix)[min_position[2]], "\n")

cat("Negative correlation:",
  round(cor_matrix[min_position[1], min_position[2]], 2), "\n")
```



Explanation

The strongest positive pair has the largest positive correlation coefficient. The strongest negative pair has the smallest correlation coefficient. The absolute value of each coefficient determines whether the relationship is weak, moderate, or strong.

Result

The R program identifies the strongest positive and negative pairs and displays their correlation coefficients. The exact variable names and coefficients should be recorded from the program output.

Question 3: Investigate the Deceptive Zero (4 Marks)

Aim

To study the relationship between fertiliser deviation and crop damage using linear and quadratic regression models.

Theory

A linear regression model represents a straight-line relationship:

\[y=a+bx \]

A quadratic regression model includes a squared term:

\[y=a+bx+cx^2 \]

The R-squared value measures the proportion of variation explained by a fitted model. A higher R-squared indicates a better fit to the observed data, but it does not prove causation.

R Program

```
# Calculate correlation

r <- cor(case$fert_dev, case$crop_damage)

cat("Correlation coefficient:", round(r, 3), "\n")


# Linear regression model

linear_model <- lm(crop_damage ~ fert_dev, data = case)


# Quadratic regression model

quadratic_model <- lm(

  crop_damage ~ fert_dev + I(fert_dev^2),

  data = case

)
```

```
# Display R-squared values
cat("Linear R-squared:",
   round(summary(linear_model)$r.squared, 4), "\n")

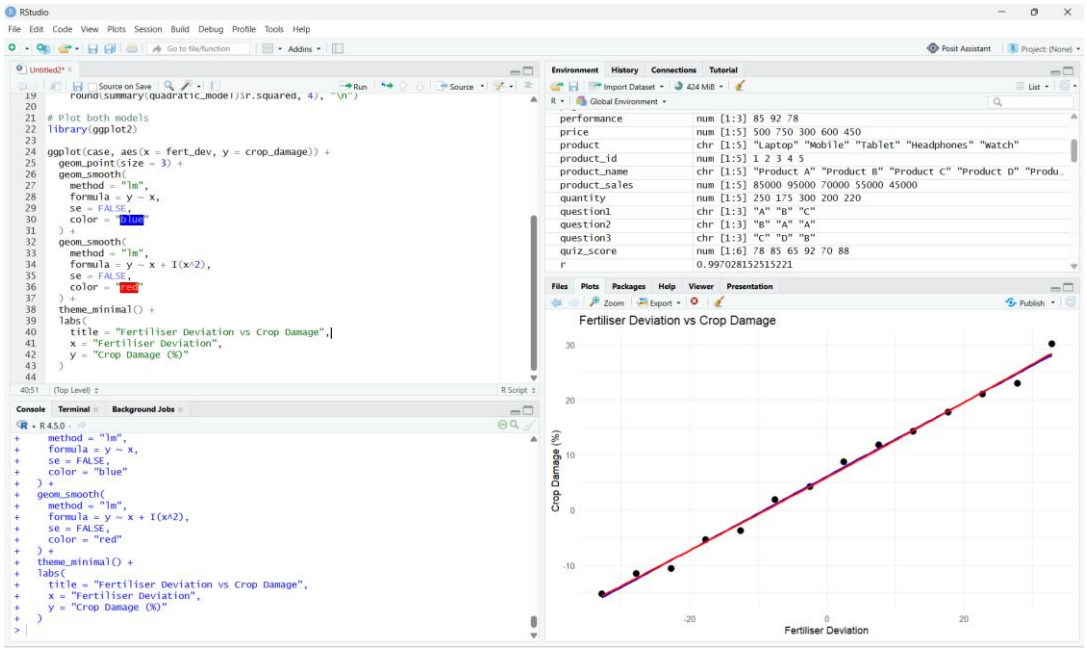
cat("Quadratic R-squared:",
   round(summary(quadratic_model)$r.squared, 4), "\n")

# Plot both models
library(ggplot2)

ggplot(case, aes(x = fert_dev, y = crop_damage)) +
 geom_point(size = 3) +
 geom_smooth(
   method = "lm",
   formula = y ~ x,
   se = FALSE,
   color = "blue"
 ) +
 geom_smooth(
   method = "lm",
   formula = y ~ x + I(x^2),
   se = FALSE,
   color = "red"
 ) +
 theme_minimal() +
 labs(
   title = "Fertiliser Deviation vs Crop Damage",
   x = "Fertiliser Deviation",
```

y = "Crop Damage (%)"

)



Explanation

The correlation coefficient measures the linear relationship between fertiliser deviation and crop damage. The linear and quadratic regression models are compared using their R-squared values. The graph displays the straight-line fit and the curved fit.

Result

The program calculates the correlation coefficient and R-squared values for both models. The results show which model explains more variation in crop damage for the observed data.

Question 4: Detect the Outlier in Fuel Cost (4 Marks)

Aim

To identify the suspicious observation in tractor hours and fuel cost and determine its effect on correlation.

Theory

An outlier is an observation that differs considerably from the other observations. It can influence the correlation coefficient and create a misleading impression of the relationship between two variables.

In the given dataset, row 5 was deliberately modified to contain 158 tractor hours and a fuel cost of 77.2 thousand rupees.

R Program

```
# Display the fifth row
cat("Suspicious record:\n")
print(case[5, ])

# Correlation using all observations
r_all <- cor(
  case$tractor_hours,
  case$fuel_cost
)

# Remove the fifth observation
case_without_row5 <- case[-5, ]

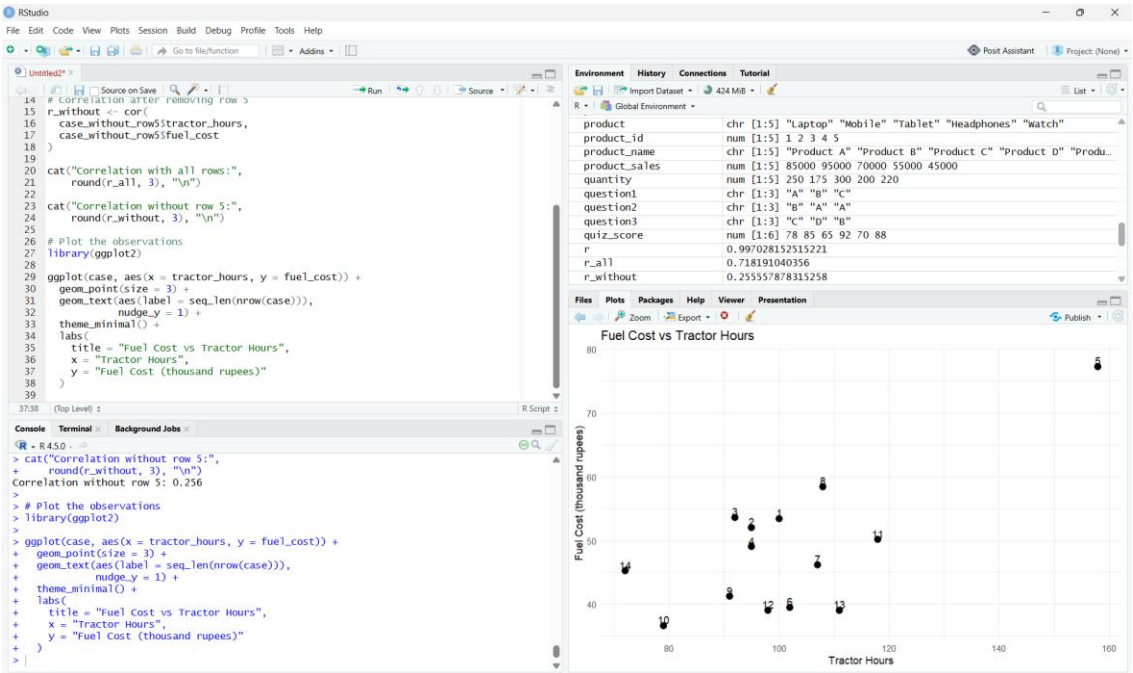
# Correlation after removing row 5
r_without <- cor(
  case_without_row5$tractor_hours,
  case_without_row5$fuel_cost
)

cat("Correlation with all rows:",
    round(r_all, 3), "\n")

cat("Correlation without row 5:",
    round(r_without, 3), "\n")

# Plot the observations
library(ggplot2)
```

```
ggplot(case, aes(x = tractor_hours, y = fuel_cost)) +
 geom_point(size = 3) +
 geom_text(aes(label = seq_len(nrow(case))),
           nudge_y = 1) +
 theme_minimal() +
 labs(
  title = "Fuel Cost vs Tractor Hours",
  x = "Tractor Hours",
  y = "Fuel Cost (thousand rupees)"
 )
```



Explanation

The correlation is first calculated using all 14 records. Then, row 5 is removed, and the correlation is calculated again using the remaining 13 records. Comparing the two values helps identify whether the unusual record has a strong influence on the relationship.

Result

Row 5 is the suspicious record because it contains 158 tractor hours and 77.2 thousand rupees in fuel cost. The two correlation coefficients show how the unusual record affects the relationship between tractor hours and fuel cost.

Question 5: Investigate Whether Grass Height Causes Milk Yield (5 Marks)

Aim

To examine the relationship between grass height and milk yield and determine whether rainfall may influence their association.

Theory

Correlation between two variables does not prove that one variable causes the other. A third variable may influence both variables and create a misleading relationship.

Partial correlation measures the relationship between two variables after controlling for another variable. In this question, rainfall is controlled because it may be related to both grass height and milk yield.

R Program

```
# Calculate three correlations
r_grass_milk <- cor(
  case$grass_height,
  case$milk_yield
)

r_rain_grass <- cor(
  case$rainfall,
  case$grass_height
)

r_rain_milk <- cor(
  case$rainfall,
  case$milk_yield
)
```

```
cat("Grass height and milk yield:",
   round(r_grass_milk, 3), "\n")

cat("Rainfall and grass height:",
   round(r_rain_grass, 3), "\n")

cat("Rainfall and milk yield:",
   round(r_rain_milk, 3), "\n")

# Regression model for grass height
grass_model <- lm(
  grass_height ~ rainfall,
  data = case
)

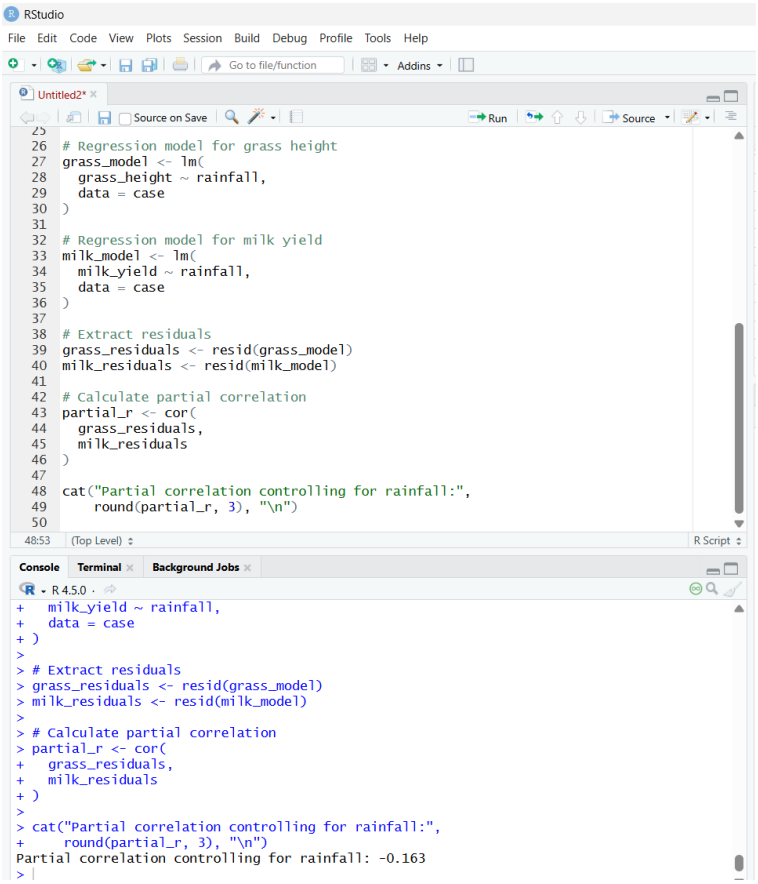
# Regression model for milk yield
milk_model <- lm(
  milk_yield ~ rainfall,
  data = case
)

# Extract residuals
grass_residuals <- resid(grass_model)
milk_residuals <- resid(milk_model)

# Calculate partial correlation
partial_r <- cor(
```

```
 grass_residuals,
 milk_residuals
)

cat("Partial correlation controlling for rainfall:",
   round(partial_r, 3), "\n")
```



Explanation

First, three correlation coefficients are calculated to examine the relationships among grass height, milk yield, and rainfall. Next, rainfall is used in two regression models, and the residuals are extracted. The correlation between these residuals estimates the partial correlation between grass height and milk yield after controlling for rainfall.

Two-Sentence Verdict

Grass height and milk yield may be correlated, but rainfall can influence both variables, so correlation alone does not prove that grass height causes an increase in milk yield. A controlled feeding experiment in which cows receive different amounts of grass under similar conditions could help test whether grass availability causes changes in milk production.

Result

The program calculates the three correlation coefficients and the partial correlation after controlling for rainfall. These results help assess whether the relationship between grass height and milk yield remains after accounting for rainfall, but they cannot independently establish causation.

Overall Conclusion

The dairy farm case demonstrates how R can be used to analyse correlations, visualise relationships, compare regression models, and detect unusual observations. The analysis also shows the importance of checking outliers and controlling for possible influencing variables before interpreting relationships. Most importantly, correlation identifies association but does not prove causation.